Shivvrat **Arya**

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Professional Summary

I am an Assistant Professor of Computer Science at the New Jersey Institute of Technology (NJIT). My research advances trustworthy, interpretable, and efficient artificial intelligence through methods that integrate neural learning, symbolic and probabilistic reasoning, and optimization. I develop algorithms for reliable decision-making, scalable inference, and structured reasoning, with applications in computer vision, video understanding, multimodal learning, and human-centered AI. My work spans neurosymbolic AI, tractable and approximate probabilistic inference, neural combinatorial optimization, explainable AI, graph-structured decision-making, and foundation models for reasoning across language and vision. My research has appeared at NeurIPS, AAAI, AISTATS, CIKM, ACM TiiS, and related venues, and has been recognized with a NeurIPS Spotlight, AAAI Oral Presentation, and TPM Best Paper Award. I have contributed to research supported by DARPA, NSF, and AFOSR.

Education

Ph.D. in Computer Science *May 2025*

The University of Texas at Dallas, Richardson, Texas *GPA: 4.0/4.0*

- **Dissertation Title:** Neural Solvers for Fast, Accurate Probabilistic Inference
- **Advisor:** Dr. Vibhav Gogate
- **Co-Advisor:** Dr. Yu Xiang

M.S. in Computer Science *May 2025*

The University of Texas at Dallas, Richardson, Texas *GPA: 4.0/4.0*

B. Tech. in Computer Science and Engineering *May 2019*

Indian Institute of Information Technology Vadodara, Vadodara, India *GPA: 8.39/10.0*

Research Areas

Artificial Intelligence (AI); Machine Learning (ML); Deep Learning (DL); Neuro-Symbolic and Trustworthy AI; Neural Combinatorial Optimization; Deep Reinforcement Learning (DRL) for Optimization and Control; Tractable and Approximate Probabilistic Inference; Scalable and Adaptive Decision-Making; Explainable and Interpretable AI (XAI/IAI); Generative and Foundation Models for Reasoning; Computer Vision and Video Understanding.

Academic Appointments and Research Experience

Department of Computer Science, Ying Wu College of Computing, NJIT *Newark, NJ*

Assistant Professor *September 2025 – Present*

- Establishing an independent research program in trustworthy and efficient artificial intelligence, with emphasis on neurosymbolic AI, tractable inference, optimization, and foundation models for reasoning.
- Developing and teaching undergraduate honors and graduate courses in artificial intelligence and machine learning, including new offerings that strengthen the department's AI curriculum.
- Advising and mentoring students through research supervision and committee service, contributing to departmental curriculum and research service, and building research collaborations.

Center for Machine Learning, The University of Texas at Dallas *Richardson, TX*

Research Assistant *August 2021 - May 2025*

- Developed neural and probabilistic inference methods for tractable models, probabilistic graphical models, and neurosymbolic systems in collaboration with faculty and research teams.
- Published work in leading AI/ML venues including NeurIPS, AAAI, AISTATS, and ACM TiiS, with research recognized through spotlight, oral presentation, and best paper honors.
- Contributed as a Graduate Research Assistant / research contributor to externally funded research programs supported by DARPA, NSF, and AFOSR.

Department of Computer Science, The University of Texas at Dallas *Richardson, TX*

Teaching Assistant *August 2020 - August 2021*

- **Courses:** Statistical Methods in AI and Machine Learning, Database Systems, Discrete Mathematics for Computing II
- **Responsibilities:** Graded assignments, held office hours, conducted project demonstrations, and developed supplemen-

tary materials and homework solutions for the courses.

Research Funding and Sponsored Support

Industry-Sponsored Research and Gifts

Lambda Research Grant Program

Principal Investigator — *2025-09-06*

- Awarded; support: $1,000 in cloud computing credits.

Prior Externally Funded Research Contributions

DARPA Perceptually-enabled Task Guidance (PTG) program — *DARPA*

Graduate Research Assistant — *Aug 2021 - May 2025*

- Advanced neurosymbolic dynamic models combining structured reasoning with deep learning.
- Improved user performance by expanding skillsets and reducing error rates in task guidance settings.
- Built robust pipelines for perception, inference, and feedback loops.

DARPA Explainable Artificial Intelligence (XAI) program — *DARPA*

Graduate Research Assistant — *Aug 2020 - Aug 2021*

- Delivered high-performance explainable models without sacrificing accuracy.
- Designed methods to increase transparency and user trust for decision support.

DARPA Assured Neuro Symbolic Learning and Reasoning (ANSR) program — *DARPA*

Graduate Research Assistant — *Aug 2023 - May 2025*

- Engineered hybrid AI algorithms integrating symbolic reasoning with data-driven learning.
- Emphasized robustness, assurance, and trustworthy deployment.

NSF Grant IIS-1652835 — *NSF*

Graduate Research Assistant — *2021 - 2025*

- Co-developed algorithms for scalable inference in graphical models.
- Supported publications recognized through best paper, spotlight, and oral presentations at NeurIPS and AAAI.

AFOSR grant FA955021S0001 (CFDA 12.800) — *AFOSR*

Graduate Research Assistant — *2021 - 2025*

- Supported research on probabilistic logic and credibility modeling for multimodal learning.
- Advanced reliable learning methods designed to reduce label requirements.

Professional Recognition and Honors

Paper Awards and Selective Presentations

*”A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models” selected as **Spotlight: top 2% papers***	**NeurIPS 2024**
***Best Paper Award** for ”A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models”*	**TPM 2024**
*”Neural Network Approximators for Marginal MAP in Probabilistic Circuits” selected for **oral presentation: top 3% papers***	**AAAI 2024**

Reviewing Recognition

*Recognized as a **Silver Reviewer** for ICML 2026*	**ICML 2026**
*Recognized as **Top Reviewer** (among the **top 8%** of reviewers) for NeurIPS 2024*	**NeurIPS 2024**

Academic Honors and Fellowships

Certificate of Academic Excellence (PhD): Awarded for Outstanding Academic Performance in the Computer Science Graduate PhD Program	**UTD**
Graduate Teaching Certificate: Awarded by the Center for Teaching and Learning	**UTD**

Certificate of Academic Excellence (MS): Awarded for Outstanding Academic Performance in the Computer Science Graduate Masters Program	**UTD**
Awarded the Jonsson School Graduate Study Scholarship (2019)	**UTD**
Awarded the Central Sector Scheme of Scholarships for College and University Students (2015), covering the full duration of Undergraduate and Graduate studies	**Department of Higher Education, India**

Publications

Peer-Reviewed Conference Publications

1. **Shivvrat Arya**, Smita Ghosh, Bryan Maruyama, Venkatesh Srinivasan, "RELINK: Edge Activation for Closed Network Influence Maximization via Deep Reinforcement Learning", *34th ACM International Conference on Information and Knowledge Management*, **CIKM**, 2025
2. Brij Malhotra, **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "Learning to Condition: A Neural Heuristic for Scalable MPE Inference", *39th Annual Conference on Neural Information Processing Systems*, **NeurIPS**, 2025
3. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "SINE: Scalable MPE Inference for Probabilistic Graphical Models using Advanced Neural Embeddings", *28th International Conference on Artificial Intelligence and Statistics*, **AISTATS**, 2025
4. Rohith Peddi, **Shivvrat Arya**, Bharath Challa, Likhitha Pallapothula, Akshay Vyas, Bhavya Gouripeddi, Qifan Zhang, Jikai Wang, Vasundhara Komaragiri, Eric Ragan, Nicholas Ruozzi, Yu Xiang, Vibhav Gogate, "CaptainCook4D: A Dataset for Understanding Errors in Procedural Activities", *The Thirty-eight Conference on Neural Information Processing Systems, Datasets and Benchmarks Track*, **NeurIPS: D&B**, 2024
5. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models", *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, **NeurIPS Spotlight: top 2% papers**, 2024
6. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "Neural Network Approximators for Marginal MAP in Probabilistic Circuits", *The 38th Annual AAAI Conference on Artificial Intelligence*, **AAAI Oral Presentation: top 3% papers**, 2024
7. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "Learning to Solve the Constrained Most Probable Explanation Task in Probabilistic Graphical Models", *27th International Conference on Artificial Intelligence and Statistics*, **AISTATS**, 2024
8. **Shivvrat Arya**, Yu Xiang, Vibhav Gogate, "Deep Dependency Networks and Advanced Inference Schemes for Multi-Label Classification", *27th International Conference on Artificial Intelligence and Statistics*, **AISTATS**, 2024
9. Vikas Chauhan, Aruna Tiwari, **Shivvrat Arya**, "Multi-Label classifier based on Kernel Random Vector Functional Link Network", *International Joint Conference on Neural Networks*, **IJCNN**, 2020

Peer-Reviewed Journal Publications

1. Reza Shahriari, Amal Hashky, **Shivvrat Arya**, Tyler Audino, Eric D. Ragan, Vibhav Gogate, Jaime Ruiz, "Comparison of Text-Based Inputs for Human-in-the-Loop Feedback in Vision-Language Models", *ACM Transactions on Interactive Intelligent Systems*, **TiiS**, 2026
2. Chiradeep Roy, Mahsan Nourani, **Shivvrat Arya**, Mahesh Shanbhag, Tahrima Rahman, Eric D. Ragan, Nicholas Ruozzi, Vibhav Gogate, "Explainable Activity Recognition in Videos using Deep Learning and Tractable Probabilistic Models", *ACM Transactions on Interactive Intelligent Systems*, **TiiS**, 2023

Peer-Reviewed Workshop Publications

1. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "Neural Network Approximators for Marginal MAP in Probabilistic Circuits", *UAI Tractable Probabilistic Modeling*, **TPM**, 2024
2. **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "A Neural Network Approach for Efficiently Answering Most Probable Explanation Queries in Probabilistic Models", *UAI Tractable Probabilistic Modeling*, **TPM: Best Paper Award**, 2024
3. Benjamin Rheault, **Shivvrat Arya**, Akshay Vyas, Jikai Wang, Rohith Peddi, Brett Benda, Vibhav Gogate, Nicholas Ruozzi, Yu Xiang, Eric D Ragan, "Predictive Task Guidance with Artificial Intelligence in Augmented Reality", *IEEE Conference on Virtual Reality and 3D User Interfaces*, **IEEE VR**, 2024
4. Rohith Peddi, **Shivvrat Arya**, Bharath Challa, Likhitha Pallapothula, Akshay Vyas, Qifan Zhang, Jikai Wang, Vasundhara Komaragiri, Nicholas Ruozzi, Eric Ragan, Yu Xiang, Vibhav Gogate, "Put on your detective hat: What's wrong in this video?", *DMLR Data-centric Machine Learning Research*, **DMLR Workshop**, 2023

Preprints and Manuscripts Under Review

1. Yeqing Chen, Cong Qi, Hanzhang Fang, Feiyang Luan, Zhirong Zhang, **Shivvrat Arya**, Zhi Wei, "CoLa-VAE: A Cell-Cell Communication-Aware Variational Autoencoder for Representation Learning and Expression Denoising", *bioRxiv*, **Preprint**, 2026
2. Brij Malhotra, **Shivvrat Arya**, Tahrima Rahman, Vibhav Gogate, "Learning to Guide Local Search for MPE Inference in Probabilistic Graphical Models", *Preprint*, **Under Review**, 2026

Advisement

Masters Advisement

NJIT

- Somana Veeramaneni (CS 700B - Master's Project).

BS Research Credits Advisement

NJIT

- Ali Elbeyali (CS 488 - Independent Study).
- Rohit Karnik (CS 488 - Independent Study; CS 489 - Computer Science Research Project).
- Shivansh Dutta (CS 489 - Computer Science Research Project).

Committee Membership

Doctoral Committee

COMMITTEE MEMBER

- Yuanjie Zou, Title: Towards Better Representations in Single-Cell Foundation Models (Committee Member).
- Shameen Shrestha, Title: The Attribution Jacobian for decomposition of evidence in an image (Qualifying Examination Committee Member).
- Xun Song, Title: MarkAug: Out-of-Distribution Detection in Single-Cell Annotation via Marker-Gene Augmentation (Qualifying Examination Committee Member).
- Mohith Oduru, Title: Spatially-Indexed Persistent Memory: A 3D-Indexed Latent Memory Module for Unsupervised Egocentric World Models (Qualifying Examination Committee Member).
- Xiaoliang Wu, Title: Evaluating Reasoning-Intensive Retrieval through Information-Seeking and Causal Attribution in Stock Price Dynamics (Qualifying Examination Committee Member).
- Hanzhang Fang, Title: Generative Spatial Deconvolution via Disentangled Latent Factors and Domain-Adaptive Decoder Transfer (Committee Member).
- Yeqing Chen, Title: Cell-Cell Communication-aware Variational Autoencoder with Dynamic Graph Laplacian Constraints (Committee Member).

Master's Thesis Committee

COMMITTEE MEMBER

- Kadir Altunel, Title: Edge Co-occurrence Regularization for Node Classification (Thesis Committee Member).
- Sabdha Ambati, Title: Benchmarking AI Understanding of Adolescent Digital Language (Thesis Committee Member).

Research Mentoring

Department of Computer Science, The University of Texas at Dallas *Richardson, TX*

RESEARCH MENTOR *August 2022 - May 2025*

- **Graduate Level:** Guided Ph.D. and Master's students in artificial intelligence, machine learning, and computer vision research, resulting in conference and journal publications, as well as successful thesis completions.
- **Undergraduate Level:** Mentored undergraduate students in artificial intelligence, machine learning, and computer vision research projects.
- **Responsibilities:** Provided technical guidance, reviewed research manuscripts, conducted regular progress meetings, and assisted in experimental design and implementation.

Classroom Teaching

Department of Computer Science, New Jersey Institute of Technology *Newark, NJ*

COURSE INSTRUCTOR *August 2025 - May 2026*

- **CS 370: Introduction to Artificial Intelligence - Honors:** Taught the honors undergraduate AI section as the sole instructor.
- **CS 785 ST: NeuroSymbolic AI:** Developed and taught a new Ph.D. seminar course expanding the department's AI curriculum; planned adaptation of course material for inclusion in the undergraduate CS curriculum.

Department of Computer Science, The University of Texas at Dallas *Richardson, TX*

GUEST LECTURER *January 2024 - December 2024*

- **Courses:** Machine Learning (Graduate Level); Artificial Intelligence (Graduate Level); Artificial Intelligence (Senior Level)
- **Responsibilities:** Delivered instructional content and moderated discussions on advanced Machine Learning concepts; Taught lectures and guided class discussions on advanced topics in Artificial Intelligence; Delivered lectures and facilitated class activities focused on core topics in Artificial Intelligence

Department of Computer Science, The University of Texas at Dallas *Richardson, TX*

TEACHING ASSISTANT *August 2020 - August 2021*

- **Courses:** Statistical Methods in AI and Machine Learning, Database Systems, Discrete Mathematics for Computing II
- **Responsibilities:** Graded assignments; Held office hours; Conducted project demonstrations; Developed supplementary materials and homework solutions

Summer School Programs *India*

HIGH SCHOOL TEACHER *Summer 2015, 2016, 2017*

- **Earlier Teaching Experience:** Taught computer science and mathematics in summer school programs in India (2015, 2016, 2017).

Certificates

Center for Teaching and Learning, The University of Texas at Dallas *Richardson, TX*

GRADUATE TEACHING CERTIFICATE RECIPIENT *2025*

- Completed formal training in instructional pedagogy through the UTD Center for Teaching and Learning.
- Fulfilled four core components: Epigeum coursework, faculty teaching observation, reflective essays, and semester-long teaching experience.
- Certificate awarded upon demonstrating teaching effectiveness and commitment to professional development in higher education.

Talks and Presentations

Invited Talks

NJIT Informatics Seminar *Newark, NJ, USA*

KNOWING WHAT COMES NEXT AND WHAT WENT WRONG: HUMAN-CENTERED AI FOR AR PROCEDURAL GUIDANCE *Mar 2026*

Santa Clara University *Santa Clara, CA, USA*

NEURAL SOLVERS FOR FAST, ACCURATE PROBABILISTIC INFERENCE *Jan 2026*

Award Talks and Presentations

The 7th Workshop on Tractable Probabilistic Modeling (TPM) *Barcelona, Spain*

A NEURAL NETWORK APPROACH FOR EFFICIENTLY ANSWERING MOST PROBABLE EXPLANATION QUERIES IN PROBABILISTIC MODELS (BEST PAPER AWARD) *May 2024*

The 38th Annual AAAI Conference on Artificial Intelligence *Vancouver, Canada*

NEURAL NETWORK APPROXIMATORS FOR MARGINAL MAP IN PROBABILISTIC CIRCUITS (ORAL PRESENTATION) *Mar 2024*

Professional Service

I regularly review papers for the following venues:

Journal Reviewer

Artificial Intelligence **(AIJ)**
Transactions on Machine Learning Research **(TMLR)**
IEEE Robotics and Automation Letters **(RA-L)**
Data Mining and Knowledge Discovery

Conference Reviewer

Neural Information Processing Systems **(NeurIPS: Recognized as a Top Reviewer in 2024)**
International Conference on Machine Learning **(ICML: Recognized as a Silver Reviewer in 2026)**
International Conference on Learning Representations **(ICLR)**
Conference on Uncertainty in Artificial Intelligence **(UAI)**
ACM International Conference on Information and Knowledge Management **(CIKM)**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD)**
International Conference on Artificial Intelligence and Statistics **(AISTATS)**

Conference Program Committee

AAAI Conference on Artificial Intelligence **(AAAI)**

Organizational Responsibilities

Completed at New Jersey Institute of Technology, Department of Computer Science

Member	**Undergraduate Committee (Fall 2025-Present)**
Member	**AI Curriculum Committee (Fall 2025-Present)**
Member	**Research Committee (Fall 2025-Present)**
Member	**Grade Appeal Committee (Fall 2025-Present)**
Member	**AI Innovations Committee (Fall 2025-Present)**

Technical Skills

Programming Languages:	Python, R, SQL, C/C++, Java, Bash, Cython
Machine Learning:	PyTorch, TensorFlow, scikit-learn, JAX, Keras, XGBoost
Deep Learning:	CNNs, Transformers, Vision-Language Models, Representation Learning, Generative Models
Neuro-Symbolic AI:	Probabilistic Reasoning, Knowledge Graphs, Logic Programming, Semantic Parsing
Computer Vision:	OpenCV, PIL, torchvision, Detection, Segmentation, Multimodal Learning
Data Science:	NumPy, Pandas, SciPy, Matplotlib, Seaborn, Jupyter
ML Infrastructure:	Docker, Kubernetes, Git, GitHub, GitHub Actions, MLflow, Weights & Biases
Cloud and HPC:	AWS, Google Cloud Platform, Azure, SLURM, CUDA

Academic Projects

Research Software

- **NeuPI: A Library for Neural Probabilistic Inference** — **Research Software**, *2026*
  Developed a PyTorch-based library for neural probabilistic inference, supporting self-supervised neural surrogates, ITSELF test-time adaptation, and MPE, Marginal MAP, and Constrained MPE tasks across probabilistic graphical models, probabilistic circuits, and neural autoregressive models.

Completed at The University of Texas at Dallas

- **Parameter and Structure Learning Algorithms for Bayesian Networks** — **Statistical Methods in AI and ML**, *Spring 2020*
  Implemented several structure learning algorithms for Bayesian Networks, including FOD-Learn (fully observed data, known structure), POD-Learn (partially observed data, known structure), and Mixture-Random-Bayes (fully observed data, unknown structure). Conducted a comparative analysis of these algorithms based on different data and structure conditions.
- **Sampling-based Variable Elimination and Conditioning** — **Statistical Methods in AI and ML**, *Spring 2020*
  Implemented Sampling-based Variable Elimination and Conditioning algorithm for performing inference on probabilistic graphical models.
- **Learning Algorithms for Bayesian Networks** — **Machine Learning**, *Fall 2019*
  Implemented four algorithms: Independent Bayesian Networks, Tree Bayesian Networks (using Chow-Liu), Mixtures of Tree Bayesian Networks using EM, and Mixtures of Tree Bayesian Networks using Random Forests.
- **Non-Iterative Neural Networks** — **Machine Learning**, *Fall 2019*
  Implemented two non-iterative neural network models for classification and regression tasks across multiple datasets.
- **Collaborative-Filtering** — **Machine Learning**, *Fall 2019*
  Implemented collaborative filtering algorithms using a subset of the Netflix Prize movie ratings data. Evaluated model performance using Mean Absolute Error and Root Mean Squared Error metrics.
- **DART Database System** — **Database Design**, *Fall 2019*
  Designed and implemented the DART system's complete database, from EER diagram creation to relational schema design, SQL-based database construction, and query/view generation.

Completed at Indian Institute of Information Technology Vadodara, India

- **Compressed Sensing** — **Computer Vision**, *Fall 2018*
  Developed and evaluated an algorithm for multi-view tracking and 3D voxel reconstruction using 2D images.
- **Kakuro Puzzle Solver** — **Artificial Intelligence**, *Spring 2018*
  Designed and implemented a bot to solve Kakuro puzzles by applying derived rules to handle diverse puzzle instances.
- **Autoencoder for Anomaly Detection** — **Deep Learning**, *Spring 2018*
  Developed an autoencoder-based anomaly detection method, successfully tested on a credit card fraud dataset.
- **LEARN: A Programming Language** — **Compiler Design**, *Spring 2018*
  Designed and implemented a beginner-friendly programming language LEARN using yacc and lex, aiming to ease the learning curve for new programmers.
- **Fatal Disease Detector Using Twitter Data** — **IIITV Hackathon**, *Fall 2018*
  Implemented a k-means clustering algorithm to detect disease spread patterns using Twitter data.
- **SoT (Security of Things)** — **Cryptography**, *Fall 2017*
  Developed an Android application to monitor real-time environmental temperature changes with AES encryption for secure data transmission.
- **Hatsphere: E-commerce Platform** — **Software Development**, *Fall 2017*
  Developed a platform to enable traditional craftsmen to sell their products online and reach a broader market.
- **Cocktail Party Effect Algorithm** — **Speech Science and Technology**, *Fall 2017*
  Implemented an algorithm to separate human voice from noise in audio recordings.
- **Movie Recommendation System** — **Database Design**, *Spring 2016*
  Developed a user interface and managed a database system for a movie recommendation engine, utilizing PostgreSQL and C for backend management.

Certifications

Coursera Course Certificates

- **Mathematics for Machine Learning: Linear Algebra** – *Present License* E5PBMECK8B4M — Apr 2019
- **What is Data Science?** – *Present License* 5WS64BF2G2SY — Feb 2019
- **Introduction to Programming with MATLAB** – *Present License* VGPWCM8WH73K — Feb 2019
- **Deep Learning Specialization** – *Present License* J9V32CC6VTB5 — Nov 2018
- **Sequence Models** – *Present License* 5E9VFH59THG4 — Nov 2018
- **Convolutional Neural Networks** – *Present License* G888N3WXPXLN — Sep 2018
- **Python Programming Essentials** – *Present License* TZXYYXTF796L — Jun 2018
- **Structuring Machine Learning Projects** – *Present License* JJNYAQTVFUPR — May 2018
- **Neural Networks and Deep Learning** – *Present License* JSAR6KKVC5Y7 — Apr 2018
- **Improving Deep Neural Networks** – *Present License* MRVNDFNWUHPQ — Apr 2018
- **Machine Learning** – *Present License* FPWNJ39A5LWQ — Mar 2018
- **Python Data Structures** – *Present License* DS5N3NM69PQ6 — Mar 2018
- **Programming for Everybody** – *Present License* XS6H2XUBJ66U — Jan 2018

Udemy Course Certificates

- The Top 5 Machine Learning Libraries in Python — Jan 2019
- MATLAB for scientists: a beginner's course — Jan 2019

Other Certifications and Professional Development

- Graduate Teaching Certificate — Center for Teaching and Learning, The University of Texas at Dallas